IE3035 Operations Research

HW4

Q1. Write the dual for each of the following primal problems:

(a) Maximize $z = -5x_1 + 2x_2$

subject to

$$-x_1 + x_2 \leq -2$$

$$2x_1 + 3x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

(b) Minimize $z = 6x_1 + 3x_2$

subject to $6x_1 - 3x_2 + x_3 \geq 2$

$$3x_1 + 4x_2 + x_3 \geq 5$$

$$x_1, x_2, x_3 \geq 0$$

(c) Maximize $z = x_1 + x_2$

subject to

$$2x_1 + x_2 = 5$$

$$3x_1 - x_2 = 6$$

x_1, x_2 unrestricted

Q2. Three refineries with daily capacities of 6, 5, and 8 million gallons, respectively, supply three distribution areas with daily demands of 4, 8, and 7 million gallons, respectively. Gasoline is transported to the three distribution areas through a network of pipelines. The transportation cost is 10 cents per 1000 gallons per pipeline mile. Table 1 gives the mileage between the refineries and the distribution areas. Refinery 1 is not connected to distribution area 3.

(a) Construct the associated transportation model.

(b) Determine the optimum shipping schedule in the network.

TABLE 1 Mileage Chart for Problem 2

		Distribution area		
		1	2	3
Refinery	1	120	180	---
	2	300	100	80
	3	200	250	120

Q3. JoShop needs to assign 4 jobs to 4 workers. The cost of performing a job is a function of the skills of the workers. Table 2 summarizes the cost of the assignments. Worker 1 cannot do job 3 and worker 3 cannot do job4. Determine the optimal assignment using the Hungarian method.

TABLE 2 Data for Problem 5

		Job			
		1	2	3	4
Worker	1	$50	$50	---	$20
	2	$70	$40	$20	$30
	3	$90	$30	$50	---
	4	$70	$20	$60	$70